Eric Romero
CSC 422

Results for Program 1

Table:

N (Power)	N (Elements)	Seq Times Median	Par Times - 2 Cores Median	Speedup (2 cores)	Par Times - 4 cores Median	Speedup (4 cores)
2^{23}	8,388,608	1.352	0.711	1.902	0.380	3.558
2^{24}	16,777,216	2.778	1.450	1.916	0.769	3.612
2^{25}	33,554,432	5.748	2.938	1.956	1.633	3.520
2^{26}	67,108,864	11.814	6.250	1.890	3.275	3.607
2^{27}	134,217,728	24.313	12.584	1.932	6.835	3.557

For each input size N (from 2^{23} to 2^{27}) I ran the program 3 times with 1 thread (sequential), 2 threads, and 4 threads, and reported the median time in seconds to keep random data from skewing the results. Speedup is calculated as Sequential time / Parallel time. Overall the parallel version scales pretty consistently: about ~1.9× faster on 2 threads and ~3.5 times faster on 4 threads, with the remaining gap from “perfect” scaling coming from thread/barrier overhead and shared-memory bandwidth limits.